

PATOJU VYSHNAVI

9949967923 | vyshnavipatoju@gmail.com

<https://www.linkedin.com/in/patoju-vyshnavi2005> | <https://github.com/Vyshu-jessy>

PROFESSIONAL SUMMARY

Computer Science graduate (B.Tech, 2026) with proficiency in Java and SQL and foundational knowledge of Python, C, and JDBC. Hands-on exposure to machine learning and federated learning through an academic healthcare-monitoring project. Demonstrated leadership, presentation, and problem-solving skills through hackathon participation and team-based projects.

TECHNICAL SKILLS

Programming: Java , Python, C

Database: SQL , JDBC

Core Concepts: OOP, DBMS, Machine Learning, Federated Learning, MLP

Project Exposure: Flask, Scikit-learn, Android/Kotlin, BLE, AWS S3, AES/RSA/JWT, CKKS (Homomorphic Encryption)

EDUCATION

B.Tech in Computer Science and Engineering — Gayatri Vidya Parishad College of Engineering for Women (GVPCEW)

Graduated 2026 | Aggregate: 83%

ACADEMIC PROJECT

FedFLAEE – Privacy-Preserving Federated Learning for IoMT Healthcare

- Contributed to a privacy-preserving healthcare-monitoring framework integrating IoMT/wearable devices, edge computing, and federated learning to enable health-condition prediction without centralizing raw patient data.
- Implemented an 8-feature MLP classifier ($8 \rightarrow 64 \rightarrow 32 \rightarrow 1$ architecture) for normal/abnormal health-condition prediction, supporting distributed local training and iterative global model aggregation across federated clients.
- Evaluated four federated learning approaches (FedAvg, FedGKD, pFedMe, pFedKD) across ~80 communication rounds using accuracy, F1-score, log loss, and ROC-AUC, with a best observed ROC-AUC of ~0.945 and ~55ms real-time prediction latency during functional testing.
- Explored secure model-update handling using AES, RSA, JWT authentication, and CKKS-based homomorphic encryption; gained exposure to Python, Flask, Scikit-learn, Android/Kotlin, BLE, and AWS S3 across the system.

CERTIFICATIONS

- NPTEL – Deep Learning, IIT Ropar
- NPTEL – Data Analytics Using Python, IIT Roorkee

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- Smart India Hackathon (SIH) – College Level:** Led a team as team lead during college-level participation in the Smart India Hackathon.
- Mathematics Hackathon – College:** Independently delivered the team's final presentation to judges after teammates were unavailable, explaining the proposed solution end-to-end — demonstrating ownership, composure, and strong presentation skills under unplanned circumstances.
- NSS Member:** Participated in beach-cleaning drives, rallies, and community awareness programs as part of the National Service Scheme.

ADDITIONAL INFORMATION

Keyboard player at church for approximately 4 years — reflecting long-term commitment and discipline.